CHRISTOPHER AVINA

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SUMMARY

Results driven and adaptable engineer with hands-on experience in software development, data engineering, and machine learning. I am proficient in Python, C++, Java, and skilled in ETL pipelines, cloud computing, game development, and big data technologies. With a proven track record of solving complex problems and adapting to new technologies, I am eager to apply my technical skillset and passion for data-driven solutions to contribute to innovative projects.

TECHNICAL SKILLS

PROGRAMMING LANGUAGES: & SOFTWARE DEVELOPMENT

Python, C, C++, SQL, R, Java, Javascript, SDLC (Agile, Scrum), Git, Docker, Linux, Bash

FRAMEWORKS & LIBRARIES:

PySpark, TensorFlow, OpenCV, Flask, Django, SKLearn, MERN Stack (MongoDB, Express, React, Node.JS), NumPy, Matplotlib

DATA ENGINEERING & CLOUD:

AWS (S3, EC2, Lambda), Databricks, Snowflake, Kubernetes, Apache Spark, Airflow, PostgreSQL, MySQL, MongoDB

EMBEDDED & ENGINEERING:

ARM/RISC-V (STM32, ATMega32x, ESP32), Systems Communication (I2C, SPI, UART, BLE), PCB Design (KiCad), Circuit Design & Analysis

EXPERIENCE

Software Engineer | Starcraft 2 Flow Q-Learning AI (Independent Research) *(January 2025-Present)*

- Developing Zerg AI agents using Q-Learning and Flow Q-Learning in PySC2.
- Logging state-action-reward transitions and tracking Q-values for analysis.
- Building statistical pipelines to evaluate decision-making performance.
- Conducted hyperparameter tuning to optimize Q-table updates and learning rates.
- Designing postgame analysis scripts to identify suboptimal decisions and retrain policies.
- Evaluating convergence trends by plotting reward curves over training iterations.
- Utilizing multi-threading to parallelize environment interactions for faster training.
- Comparing exploration-exploitation efficiency using statistical metrics.

Data Engineer | NASA ELT Pipeline (Freelance) *(August 2024-Sept 2024)*

- Designed and implemented an ELT pipeline in Databricks to extract, transform, and load Mars Rover image data.
- Utilized PySpark for distributed data processing and Delta Lake for efficient storage and versioning.
- Optimized transformations with Spark SQL for querying and filtering mission-specific images.
- Automated ingestion workflows using Apache Airflow for scheduling and monitoring.
- Implemented data visualization with Matplotlib to analyze image distribution and quality.
- Applied AWS S3 for scalable data storage and retrieval.

Health Break | Cancer Recovery *(June 2023 - January 2025)*

- Successfully minimized cancer symptoms using my own protocols developed based on statistical modeling.
- Began independent research on a Starcraft 2 AI agent using Flow Q-Learning and PySC2.
- Built an ELT pipeline that extracts and transforms Mars Rover image data for analysis in Databricks.
- Built an Algorithm visualizer using C++ and SDL2
- Contributed to an open-source passion project in game development
- Completed Colt Steele's Web Development Bootcamp
- Completed building out a portfolio using the knowledge gained from the bootcamp.
- Cancer Free

Embedded Software Engineer | Rekovar *(April 2023 - June 2023)*

- Led a comprehensive overhaul of project structure and build systems, increasing engineer productivity by 10%.
- Developed and debugged neonatal medical device software, ensuring compliance with FDA Class II regulations.
- Applied unit testing best practices (Ceedling, Unity, CMock) in CI/CD workflows to ensure medical software reliability.
- Created documentation and SOPs for FDA compliance via Greenlight Guru.
- Left due to cancer treatments; now cancer-free.

EDUCATION

UNIVERSITY OF CALIFORNIA, IRVINE

B.S. Computer Science & Engineering | March 2023